Assignment 2

Q1. Trace Bubble Sort Passes

Problem Description:
Given an array, print the array after each pass of Bubble Sort.

Example:
Input: `[4, 3, 2]`

Output:
Pass 1: `[3, 2, 4]`
Pass 2: `[2, 3, 4]`

Q2. Insert an Element into a Sorted Array

Problem Description:
Given a sorted array, insert a new element into its correct position using the logic of Insertion Sort.

Example:
Input: `[1, 3, 5, 7]`, insert `4`
Output: `[1, 3, 4, 5, 7]`

Requirements:

- Do not use extra sorting.
- Shift elements as needed.

Q3.Selection Sort in Descending Order

Problem Description:
Modify the Selection Sort algorithm to sort an array in descending order.

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Example:
Input: `[3, 5, 1, 2]`
Output: `[5, 3, 2, 1]`

Q4.Optimized Bubble Sort

Problem Description:
Modify the Bubble Sort algorithm to improve its efficiency by stopping early if the array is already sorted.

Requirements:

- If no swaps happen in a pass, terminate early.